

WORKSHOP

TURING MACHINES AND DECIDABILITY – 2ND PART

- One way to show that the language E_{DFA} is decidable is to construct a Turing machine T that works as defined below.

$$E_{DFA} = \{ \langle B \rangle \mid B \text{ is a DFA and } L(B) = \emptyset \}$$

Decider D

On input $\langle B \rangle$:

- Let $R := \{ q_0 \}$ (q_0 is the initial state)
- Repeat: for every $q \in R$ and every $a \in \Sigma$, if there is a transition $\delta(q, a) = p$, add p to R
- Stop when R no longer grows
- If $R \cap F = \emptyset$ accept (language is empty); else, reject.

- Which states are in R when the loop ends in step 3?
 - Explain why if $R \cap F = \emptyset$ then $L(B)$ is an empty set.
 - Explain why D is a decider (i.e. why always halt).
 - What is the time complexity of D when the input is an automaton with Q states, T transitions, and N symbols in the alphabet?
- Show that $A_{NFA} = \{ \langle B, w \rangle \mid B \text{ is a NFA and } B \text{ accepts } w \}$ is decidable.
 - Explain, in your own words, why we know that there exists languages that are not recognizable.
 - Given the language:

$$E_{CFG} = \{ \langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset \}$$

And knowing that $A_{CFG} = \{ \langle G, w \rangle \mid G \text{ is a CFG and } G \text{ derives } w \}$ is decidable

We propose the following argument to show that E_{CFG}

PROOF IDEA:

Since A_{CFG} is decidable, there exists a TM A that decides A_{CFG}

Let be $s_1, s_2, s_3, \dots$ the strings that belong to Σ^* , where Σ is the alphabet of G

Build a TM E that “on input $\langle G \rangle$:

Enumerate $s_1, s_2, s_3, \dots$ one by one

For each s_i ($i \geq 1$) run A with input $\langle G, s_i \rangle$

If A accepts, reject

Else continue with the next string

Analyze the proof idea and answer:

- a) Does this proof idea apply the concept of reduction? If so, which problem is reduced to which other problem? Justify your answer
- b) Does this idea prove that E_{CFG} is decidable? Justify your answer

5. Given the language

$$E_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$$

And the following idea to prove decidability of E_{TM} :

We suppose E_{TM} is decidable, therefore exists a TM E that decides E_{TM}

Then, we can build a TM S that decides A_{TM} (the acceptance problem) as follows:

$S =$ “On input $\langle M, w \rangle$

- 1) Use $\langle M \rangle$ and $\langle w \rangle$ to construct a TM M'

$M' =$ “On input x : // on any input

If $x \neq w$, reject

If $x = w$, simulate M on input w and accept if M does.”

(Note: M' has M and w hardwired)

- 2) Run E on input $\langle M' \rangle$

- 3) If E accepts, reject; if E rejects, accept

Analyze the proof idea and answer:

- a) What strings could M' accept? Justify your answer
- b) What strings does M' reject? Justify your answer
- c) What is the maximum size of $L(M')$?
- d) Explain why S answers to the question: *Does M accept w ?*
- e) Does this proof idea apply the concept of reduction? If so, which problem is reduced to which other problem? Justify your answer
- f) Provide the conclusion of the proof by concluding whether E_{TM} is decidable or undecidable